

Flatcar Container Runtimes

Flatcar ships Docker and containerd as built-in sysex images. Both can be disabled and replaced with custom versions. Kubernetes is not bundled but is easily deployed via kubeadm or sysex images from `sysex-bakery`.

Docker

Docker and containerd ship as sysex images (`docker-flatcar`, `containerd-flatcar`) enabled by default since Flatcar 3794.0.0.

Getting started:

```
docker run -it ubuntu bash
docker pull nginx
```

Customizing Docker (daemon options via Butane):

```
storage:
  files:
    - path: /etc/docker/daemon.json
      contents:
        inline: |
          {
            "log-driver": "journald",
            "storage-driver": "overlay2"
          }
```

Using a custom Docker version — disable the built-in sysex and place a custom binary in `/opt/bin/` or use a custom sysex image from `sysex-bakery`. See [flatcar-provisioning](#) for sysex disable pattern.

Registry authentication — credentials in `~/.docker/config.json` or `/etc/docker/config.json` (for system-wide). Kubernetes pulls use `imagePullSecrets`.

containerd

containerd is included alongside Docker. For Kubernetes nodes it is common to use containerd directly without Docker.

Switching to containerd-only for Kubernetes:

Disable Docker sysex via Ignition (see [flatcar-provisioning](#)) and configure containerd via

/etc/containerd/config.toml .

Kubernetes

Flatcar is a first-class Kubernetes node OS. Kubernetes itself is not bundled but is straightforward to deploy.

Compatibility Matrix (as of 2026)

Kubernetes Version	Alpha	Beta	Stable	LTS (2024)
1.31–1.33	Known working	Known working	Known working	Known working
1.34–1.36	Tested	Tested	Tested	Tested

Tested CNIs: **Cilium**, **Flannel**, **Calico**

Known issue: Flannel >0.17.0 does not work with enforced SELinux ([flatcar#779](#)).

Deploying with kubeadm + sysex

The recommended approach uses the Kubernetes sysex from `sysex-bakery` and `systemd-sysupdate` for in-place Kubernetes version upgrades.

Butane config snippet (control plane with sysex):

```
variant: flatcar
version: 1.0.0
storage:
  links:
    - target: /opt/extensions/kubernetes/kubernetes-v1.33.2-x86-64.raw
      path: /etc/extensions/kubernetes.raw
  files:
    - path: /etc/sysupdate/kubernetes.d/kubernetes-v1.33.conf
      contents:
        source:
          https://extensions.flatcar.org/extensions/kubernetes/kubernetes-v1.33.conf
    - path: /opt/extensions/kubernetes/kubernetes-v1.33.2-x86-64.raw
      contents:
        source: https://extensions.flatcar.org/extensions/kubernetes-
v1.33.2-x86-64.raw
    - path: /etc/hostname
      contents:
        inline: "flatcar-node1"
```

Use [Kured](#) or [FLUO](#) for coordinated node reboots during Flatcar OS updates.

High-Availability Kubernetes

The `container-runtimes/high-availability-kubernetes` doc covers multi-control-plane setups. Key points:

- Use an external load balancer for the API server
- etcd can be stacked (on control plane nodes) or external
- FLUO handles draining and rebooting nodes safely during OS updates

Incus (LXC/VM manager)

Incus is available as an opt-in sysextr since Flatcar 4285.0.0:

```
incus
```

in `/etc/flatcar/enabled-sysextr.conf`. Provides LXD-compatible container and VM management. See `container-runtimes/incus` for setup.

Managing Container Services with systemd

Containers can be managed as systemd services rather than via Docker Compose or similar tools. This integrates them with Flatcar's update and restart semantics:

```
[Unit]
Description=My App
After=docker.service
Requires=docker.service

[Service]
ExecStart=/usr/bin/docker run --name myapp --rm myimage:latest
Restart=always

[Install]
WantedBy=multi-user.target
```

Defined via Butane `systemd.units` — see [flatcar-provisioning](#).